



YEAR 4 SEMESTER 1
EET 4304
POWER ELECTRONICS

Assignment 01

Design and Analysis of a Two-Stage PWM Controlled DC–DC Buck Converter

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1. Problem Statement

The objective of this project is to design a portable solar-powered charging system using a two-stage DC–DC buck converter. The system uses solar energy to charge a 12 V lead-acid battery and to supply a stable low-voltage output for electronic devices.

In the first stage, the converter steps down the solar panel voltage (15–20 V) to the required battery charging range of 12–14.4 V. Since solar output changes with sunlight and temperature, the converter must provide stable regulation under varying conditions to ensure safe and effective battery charging.

In the second stage, the battery voltage (12–14.4 V) is converted to a regulated 5 V, 3 A USB output suitable for small devices. This output must have good voltage regulation and low ripple.

The system is required to maintain efficiency above 85% to reduce power losses and improve reliability. Additionally, the design should use low-cost and locally available components so it can be practically implemented in real-world.

2. Design Approach & Justification

This project uses a solar-powered two-stage buck converter to convert a varying solar panel voltage into stable DC output voltages. The design focuses on high efficiency, low cost, simplicity, and ease of understanding, making it suitable for both academic learning and practical applications.

Instead of using complex SMPS controller ICs, an analogue closed-loop PWM control method is employed. A ramp signal is generated and compared with the output voltage feedback. The output

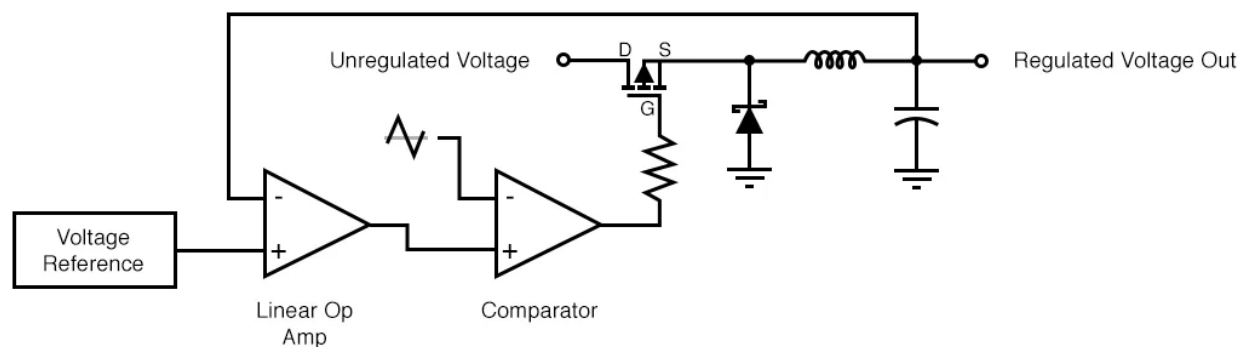


Figure 1 Closed loop PWM controlled SMPS design

voltage is sensed using a resistor divider and compared with a fixed reference voltage. If the output voltage increases above the set value, the PWM duty cycle is reduced, and if the output voltage decreases, the duty cycle is increased. This closed-loop control maintains a stable output voltage under varying input and load conditions while keeping the circuit simple and low cost using locally available components.

The NE555 timer is used in a stable mode to generate a sawtooth (ramp) waveform at a switching frequency of 100 kHz. This frequency provides a good balance between efficiency and component size, allowing smaller inductors and capacitors while keeping switching losses low. The same switching frequency is used for both converter stages to simplify the design.

A TL431 is used as a precision reference and error sensing element. A voltage divider scales the output voltage and feeds it to the TL431. Variations in output voltage change the conduction of the TL431, producing an error signal that controls the PWM duty cycle.

The LM311 comparator compares the ramp waveform with the error voltage and generates the PWM signal. The duty cycle automatically adjusts based on output voltage variations, ensuring proper regulation. The TC4420 gate driver is used to provide sufficient drive current to the MOSFET gate, enabling fast and clean switching, reducing losses, and improving efficiency.

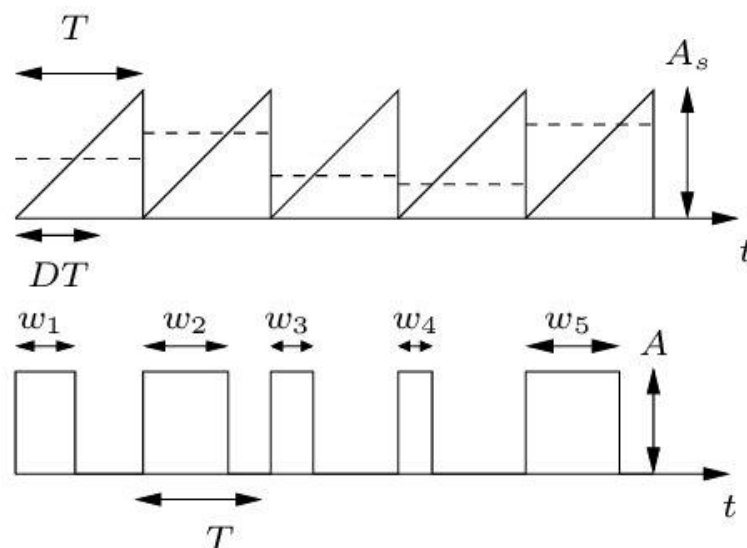


Figure 2 PWM generation using ramp and error voltage

The power stage operates as a buck converter. When the MOSFET is ON, energy is stored in the inductor, and when it is OFF, the stored energy is transferred to the load through the freewheeling diode. The output capacitor reduces voltage ripple and ensures smooth DC output. By controlling the PWM duty cycle, the converter maintains a stable output voltage even under changing solar and load conditions.

3. Calculations

3.1 Midpoint Input Voltages Used

Stage 1 input range: 15–20 V

$$V_{in1} = \frac{15 + 20}{2} = 17.5 \text{ V}$$

Stage 2 input range: 12–14.4 V

$$V_{in2} = \frac{12 + 14.4}{2} = 13.2 \text{ V}$$

3.2 Duty Cycle (Midpoint Conditions)

(ideal buck converter approximation $D = \frac{V_o}{V_{in}}$)

Stage 1 (17.5 V $\rightarrow$ 14.4 V)

$$D_1 = \frac{14.4}{17.5} = 0.823$$

Stage 2 (13.2 V $\rightarrow$ 5 V)

$$D_2 = \frac{5}{13.2} = 0.379$$

3.3 Inductor Selection (100 kHz, 30% ripple)

Given switching frequency:

$$f_s = 100 \text{ kHz}$$

Inductor ripple target:

$$\Delta I_L = 0.3 I_o$$

Inductor equation for buck converter:

$$L = \frac{(V_{in} - V_o)D}{\Delta I_L f_s}$$

Stage 1

Output current $I_{o1} = 1 \text{ A}$

$$\Delta I_{L1} = 0.3 \times 1 = 0.3 \text{ A}$$

$$L_1 = \frac{(17.5 - 14.4) \times 0.823}{0.3 \times 100000} = 85 \mu\text{H}$$

Stage 2

Output current $I_{o2} = 3 \text{ A}$

$$\Delta I_{L2} = 0.3 \times 3 = 0.9 \text{ A}$$

$$L_2 = \frac{(13.2 - 5) \times 0.379}{0.9 \times 100000} = 34.5 \mu\text{H}$$

3.4 Output Capacitor Selection

Capacitor ripple equation (triangular ripple approximation):

$$C = \frac{\Delta I_L}{8f_s \Delta V}$$

Stage 2 (50 mV ripple)

$$\Delta V_2 = 0.05 \text{ V}, \Delta$$

$$I_{L2} = 0.9 \text{ A}$$

$$C_2 = \frac{0.9}{8 \times 100000 \times 0.05} = 22.5 \mu\text{F}$$

Stage 1 (100 mV ripple)

$$\Delta V_1 = 0.1 \text{ V}, \Delta$$

$$I_{L1} = 0.3 \text{ A}$$

$$C_1 = \frac{0.3}{8 \times 100000 \times 0.1} = 3.75 \mu\text{F}$$

3.5 TL431 Divider Calculations (Correct Formula)

For TL431 in regulation:

$$V_o = V_{ref} \left(1 + \frac{R_1}{R_2} \right)$$

where $V_{ref} = 2.495 \text{ V}$

5 V output setting

Using $R_1 = 100k\Omega$,

$$R_2 = 93k\Omega:$$

$$V_o = 2.495 \left(1 + \frac{100}{93} \right) \approx 5.17 \text{ V}$$

14.4 V output setting

Using $R_1 = 10.5k\Omega$,

$R_2 = 2.2k\Omega$:

$$V_o = 2.495 \left(1 + \frac{10.5}{2.2}\right) \approx 14.4 \text{ V}$$

3.6 Power Efficiency Calculations (Both Stages)

Output power

$$P_o = V_o I_o$$

To estimate efficiency, the main losses considered are:

- MOSFET conduction loss
- Diode conduction loss
- Inductor copper loss
- MOSFET switching loss

Assumptions

- $R_{DS(on)} = 0.03 \Omega$
- Diode forward drop $V_f = 0.5 \text{ V}$
- Inductor copper resistance $R_L = 0.1 \Omega$
- Switching transition time: $(t_r + t_f) = 50 \text{ ns}$
- Switching frequency: $f_s = 100 \text{ kHz}$

Inductor RMS current

$$I_{L,rms} = \sqrt{I_o^2 + \frac{(\Delta I_L)^2}{12}}$$

MOSFET RMS current during ON time (approx.)

$$I_{sw,rms} \approx I_{L,rms} \sqrt{D}$$

Loss equations used

MOSFET conduction

$$P_{MOS} = I_{sw,rms}^2 R_{DS(on)}$$

Diode conduction (average current approx.)

$$P_D = V_f$$

$$I_o(1 - D)$$

Inductor copper loss

$$P_L = I_{L,rms}^2 R_L$$

Switching loss (common estimate)

$$P_{sw} \approx \frac{1}{2} V_{in} I_o (t_r + t_f) f_s$$

Efficiency

$$\eta = \frac{P_o}{P_o + P_{loss}}$$

Stage-1 (17.5 V $\rightarrow$ 14.4 V, 1 A)

Given:

- $P_o = 14.4 \text{ W}$
- $\Delta I_{L1} = 0.3 \text{ A}$
- $D_1 = 0.823$

Calculated:

- $I_{L,rms} \approx 1.0037 \text{ A}$
- $I_{sw,rms} \approx 0.911 \text{ A}$

Losses:

- $P_{MOS} \approx 0.0249 \text{ W}$
- $P_D \approx 0.0885 \text{ W}$
- $P_L \approx 0.1007 \text{ W}$
- $P_{sw} \approx 0.0438 \text{ W}$

Total:

- $P_{loss} \approx 0.258 \text{ W}$

Efficiency:

$$\eta_1 = \frac{14.4}{14.4 + 0.258} \approx 98.2\%$$

Stage-2 (13.2 V $\rightarrow$ 5 V, 3 A)

Given:

- $P_o = 15 \text{ W}$
- $\Delta I_{L2} = 0.9 \text{ A}$
- $D_2 = 0.379$

Calculated:

- $I_{L,rms} \approx 3.012 \text{ A}$
- $I_{sw,rms} \approx 1.855 \text{ A}$

Losses:

- $P_{MOS} \approx 0.103 \text{ W}$
- $P_D \approx 0.932 \text{ W}$
- $P_L \approx 0.454 \text{ W}$
- $P_{sw} \approx 0.099 \text{ W}$

Total:

- $P_{loss} \approx 1.588 \text{ W}$

Efficiency:

$$\eta_2 = \frac{15}{15 + 1.588} \approx 90.4\%$$

both stages satisfy the **>85% efficiency target** at midpoint operating conditions.

4. Circuit Diagrams

The following figures show the MATLAB/Simulink implementation of the two buck converter stages and the Proteus schematic used for the analog PWM control and power stage verification.

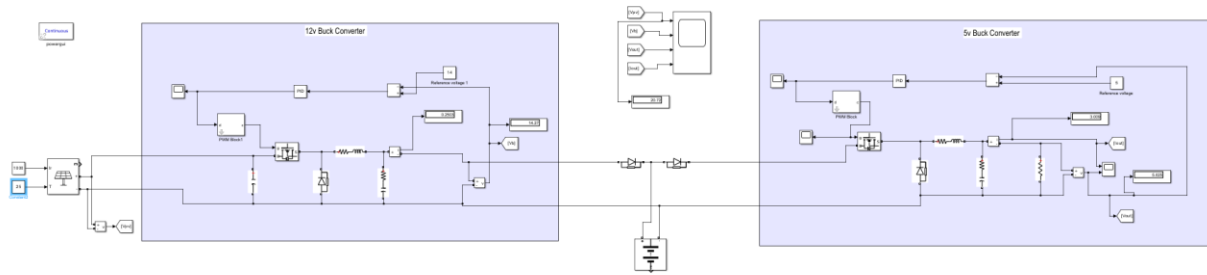


Figure 3 MATLAB/Simulink model

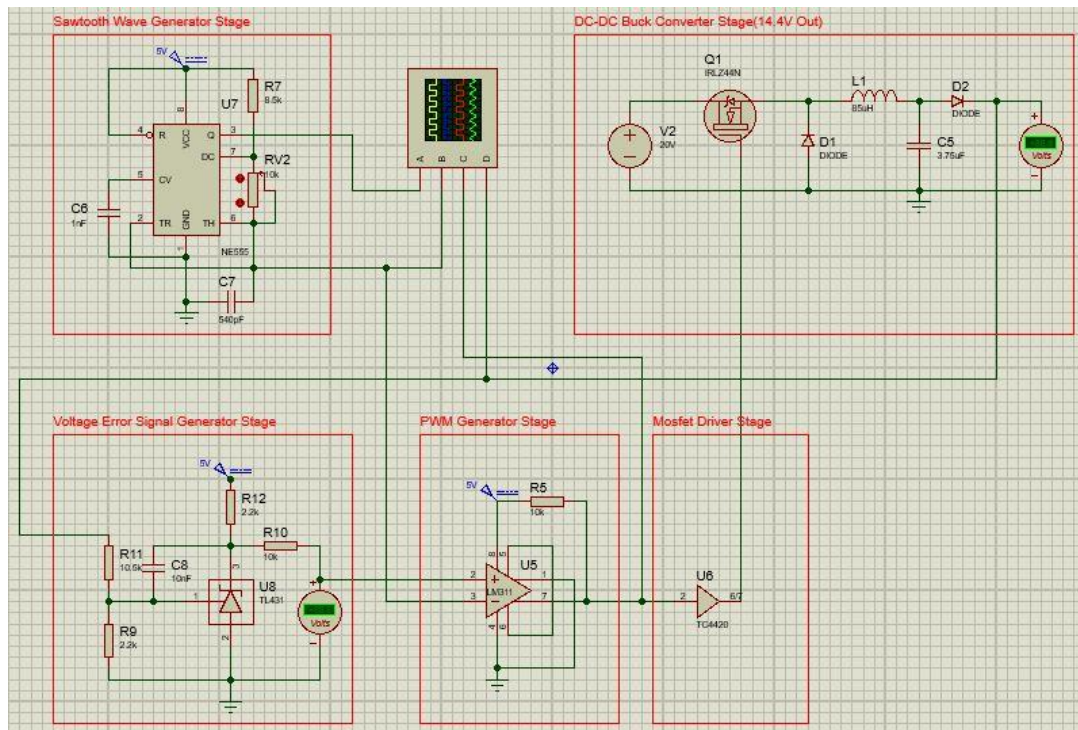


Figure 4 Proteus schematic of Stage-I analog PWM control and 14.4 V buck power stage

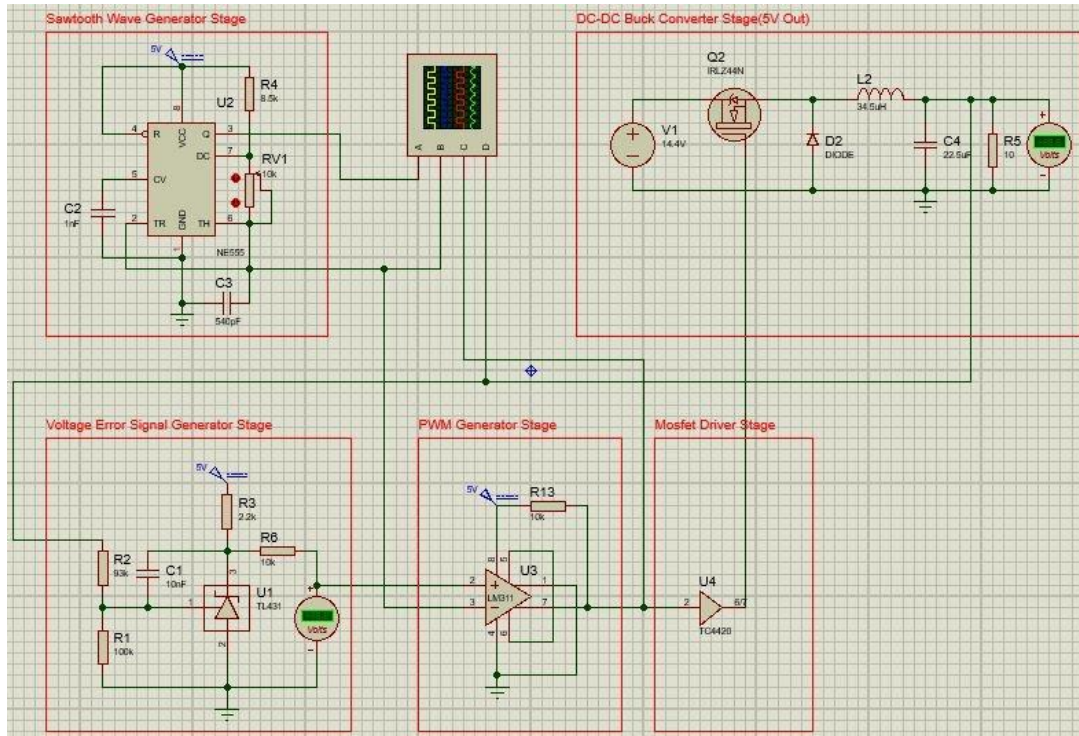


Figure 5 Proteus schematic of Stage-2 analog PWM control and 5 V buck power stage

5. Component Selection Table

Table 1 Component Selection Table

Block	Component	Selected Value/Part
Stage-1 Inductor	L1	85 μ H
Stage-2 Inductor	L2	34.5 μ H
Stage-1 Cout	C1	3.75 μ F
Stage-2 Cout	C2	22.5 μ F
Reference	TL431	2.495 V
PWM ramp	NE555	100 kHz
Comparator	LM311	PWM compare
Driver	TC4420	MOSFET driver
MOSFET	IRLZ44N	Power switch
Diode	Schottky	≥ 5 A

6. Simulation Results

6.1.MATLAB/Simulink simulations Results

6.1.1.MATLAB/Simulink simulations results under 1000 W/m² irradiance

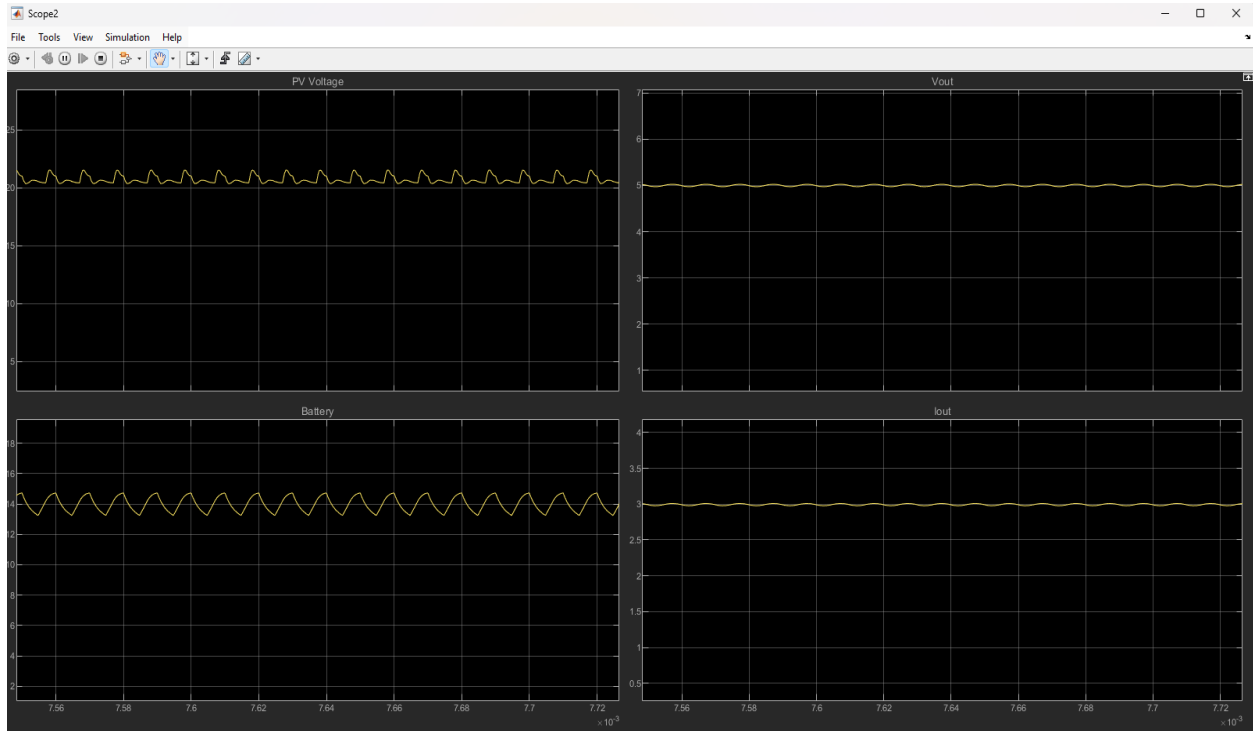


Figure 6 PV voltage, regulated 5 V output, battery voltage, and output current under 1000 W/m² irradiance.

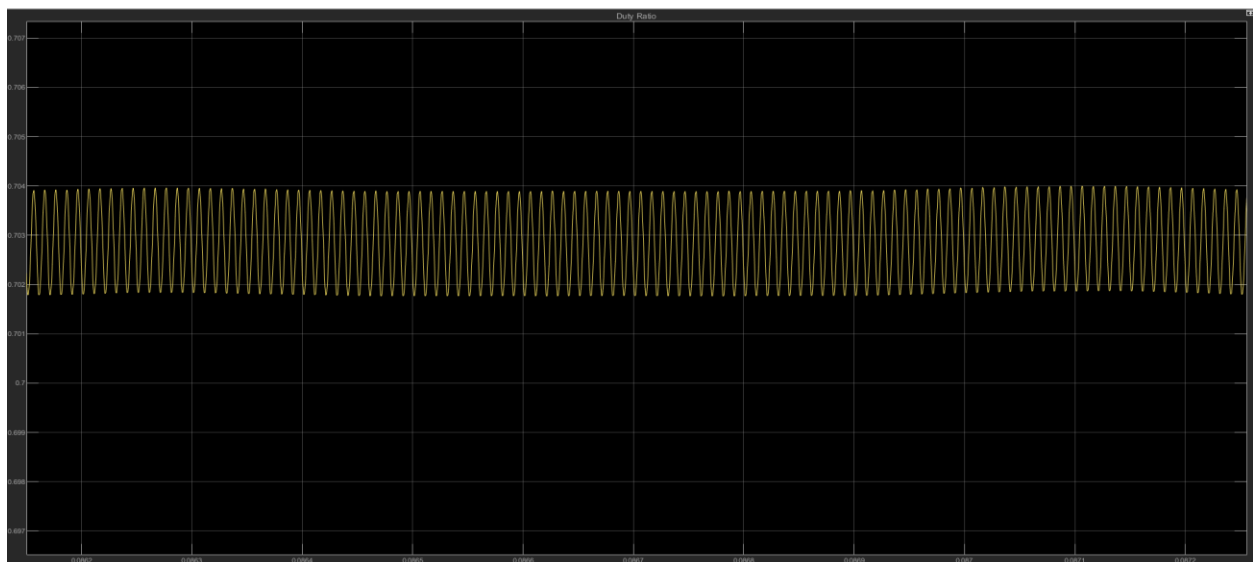


Figure 7 Duty Ratio Stage 1 under 1000 W/m² irradiance.

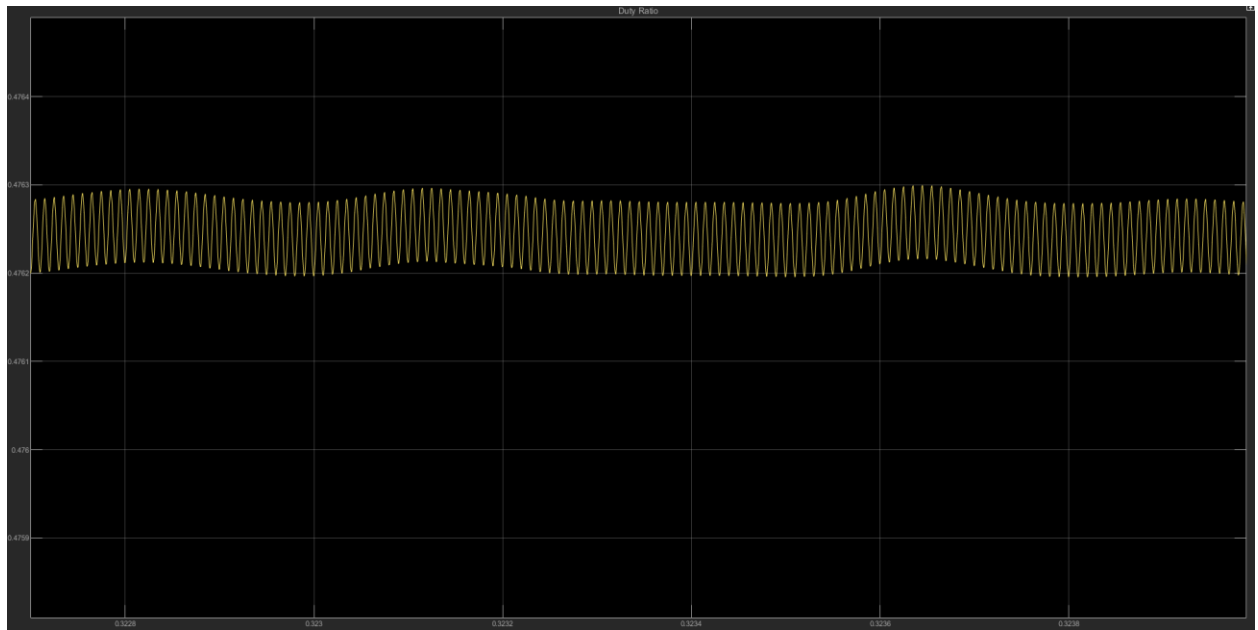


Figure 8 Duty Ratio Stage 2 under 1000 W/m² irradiance.

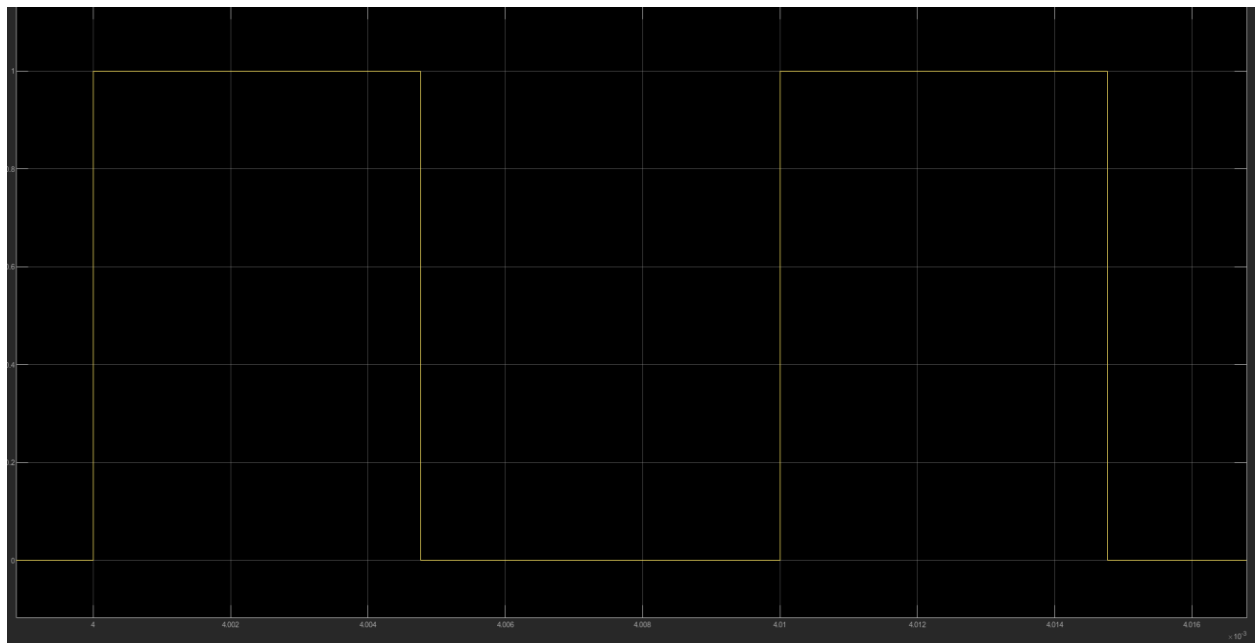


Figure 9 MOSFET gate drive / switching waveform Stage 1 under 1000 W/m² irradiance.

6.1.2.MATLAB/Simulink simulations results under 500 W/m² irradiance

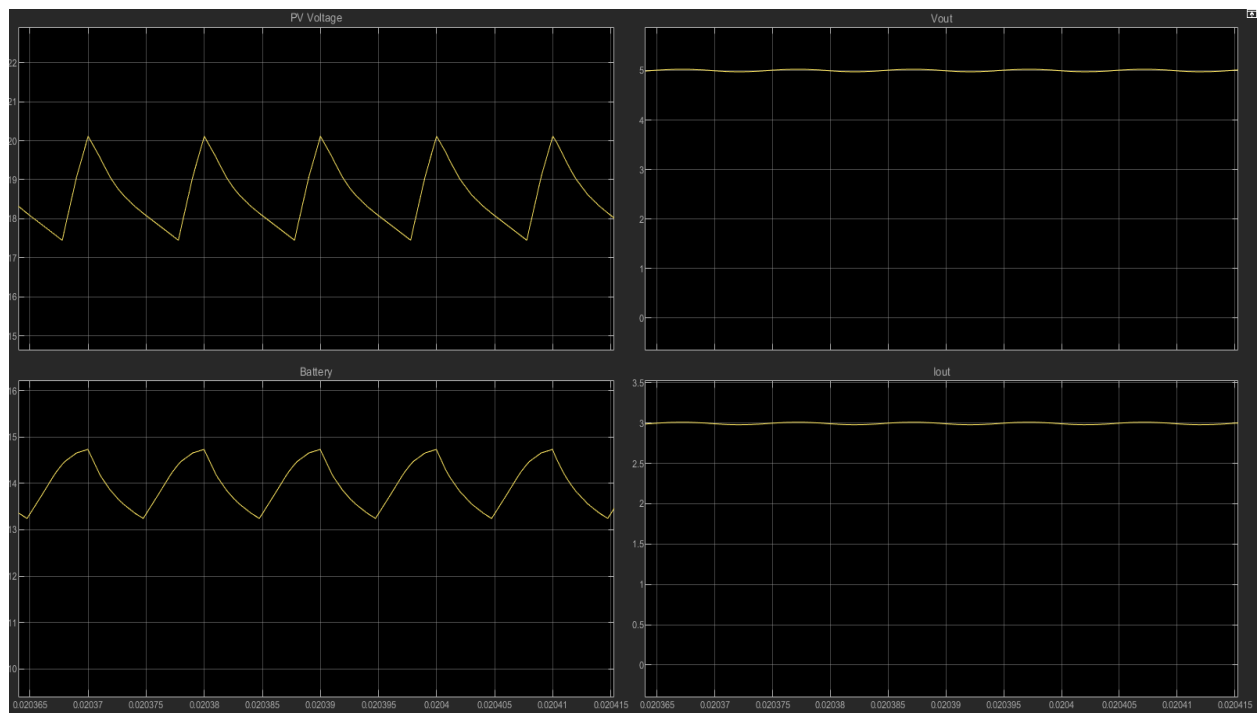


Figure 10 PV voltage, regulated 5 V output, battery voltage, and output current under 500 W/m² irradiance.

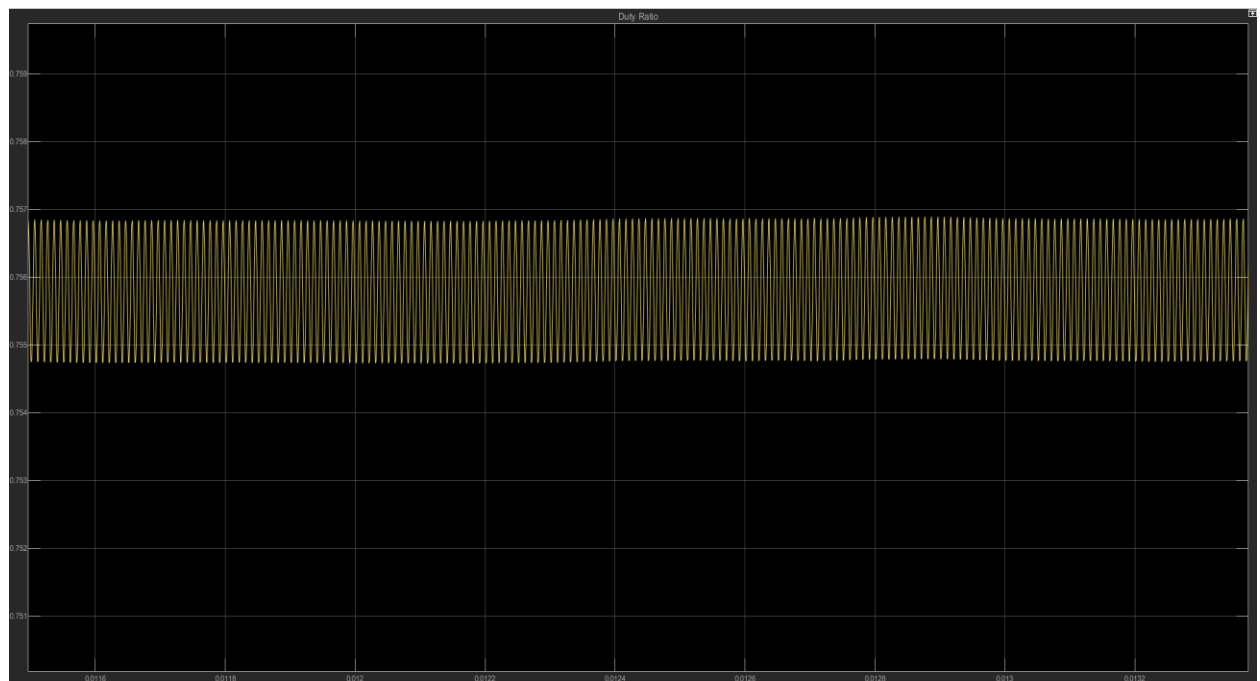


Figure 11 Duty Ratio Stage 1 under 500 W/m² irradiance.

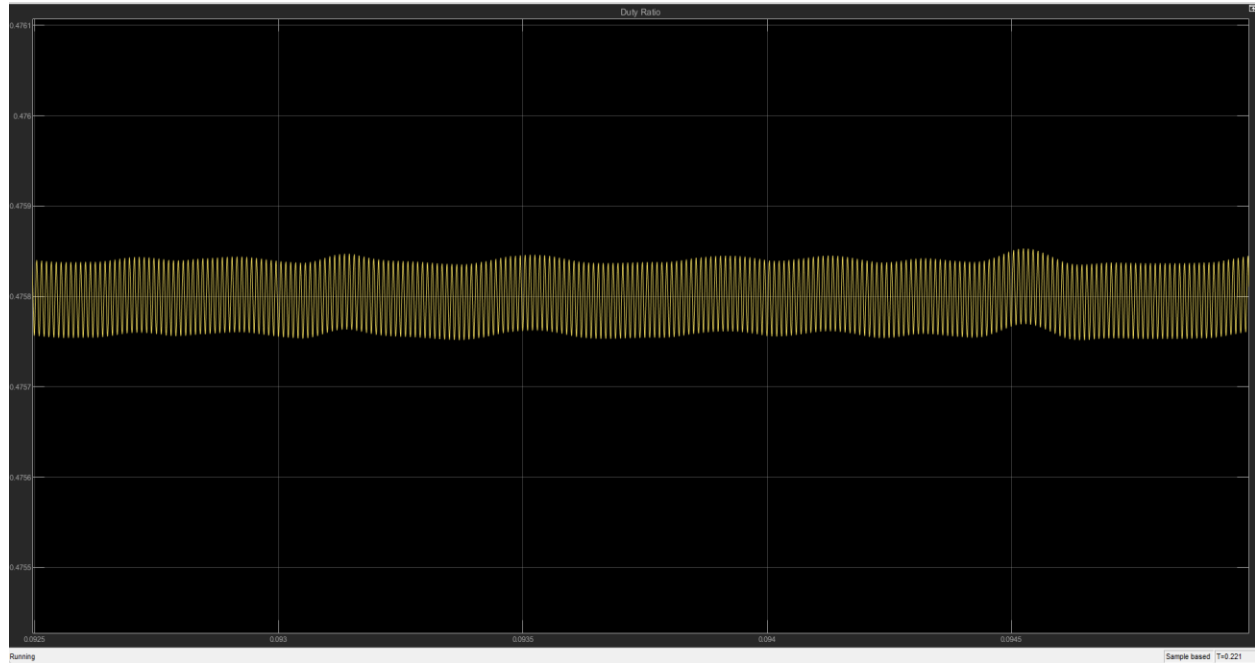


Figure 12 Figure 7 Duty Ratio Stage 2 under 500 W/m^2 irradiance.

6.2. Proteus simulations Results

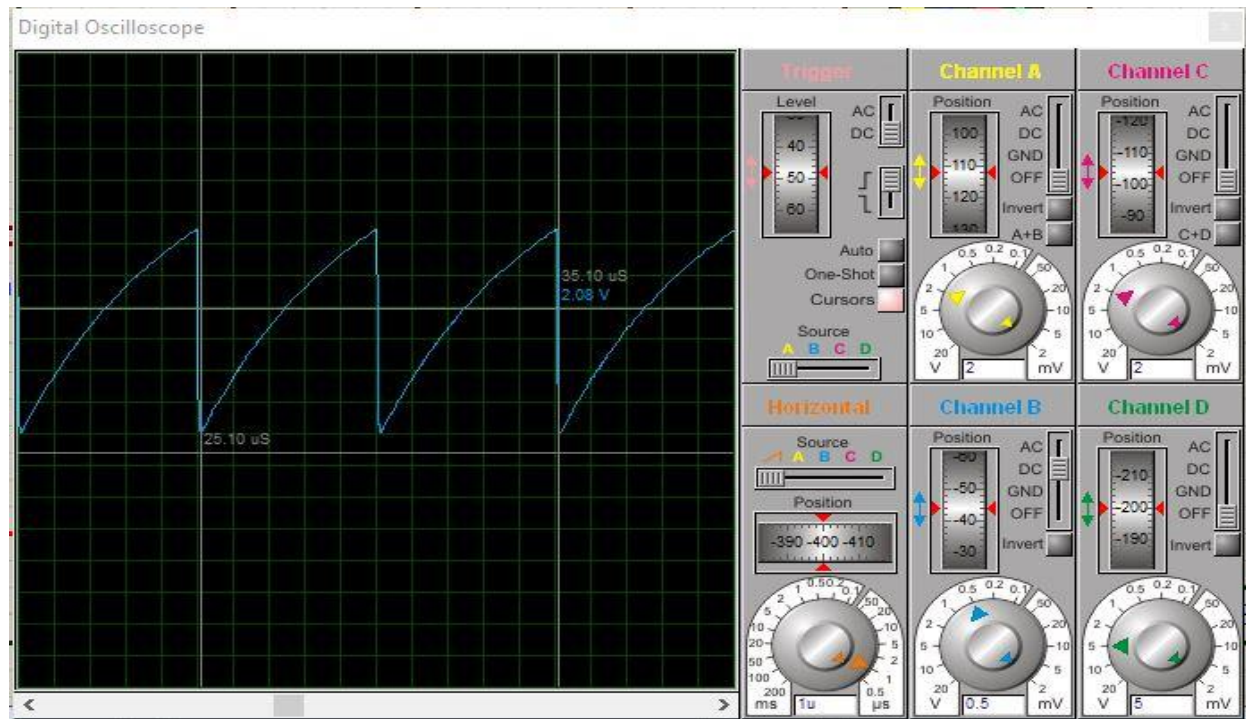


Figure 13 Figure 6: 100kHz sawtooth wave form for 14.4V and 5V

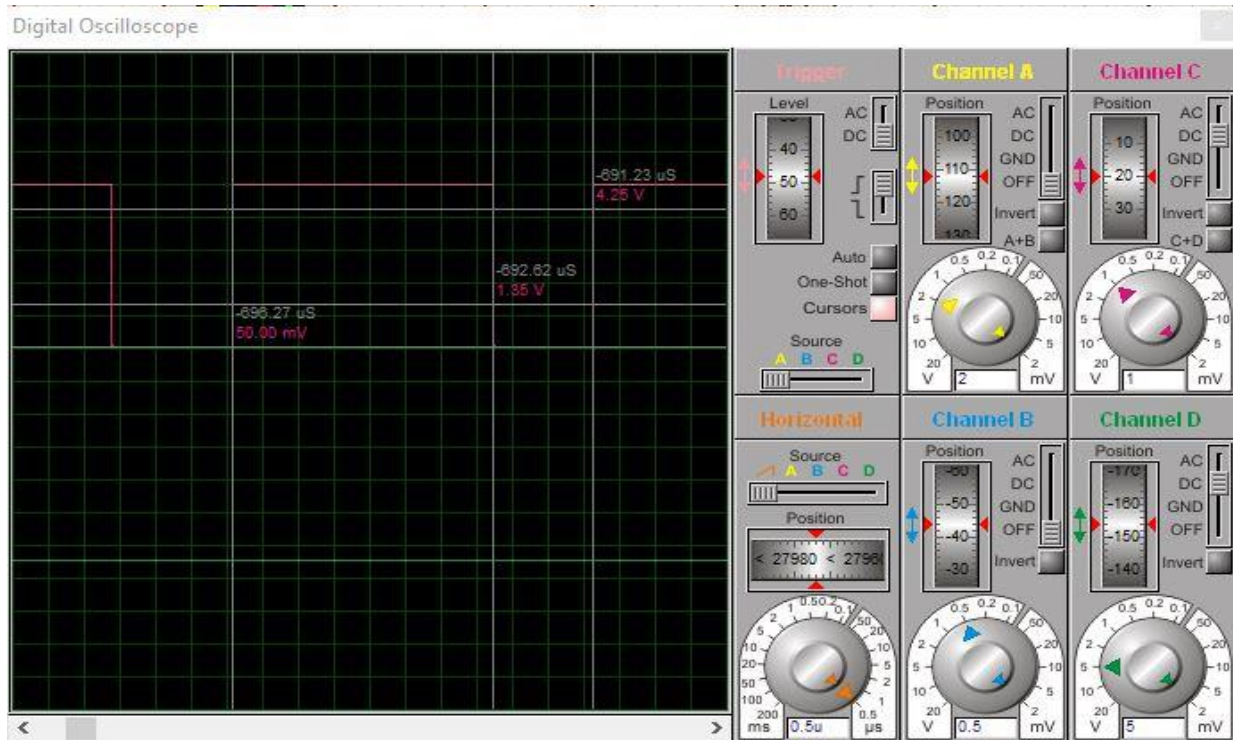


Figure 14 72.42% PWM duty cycle and 14.5V out at 20V (Stage-1)

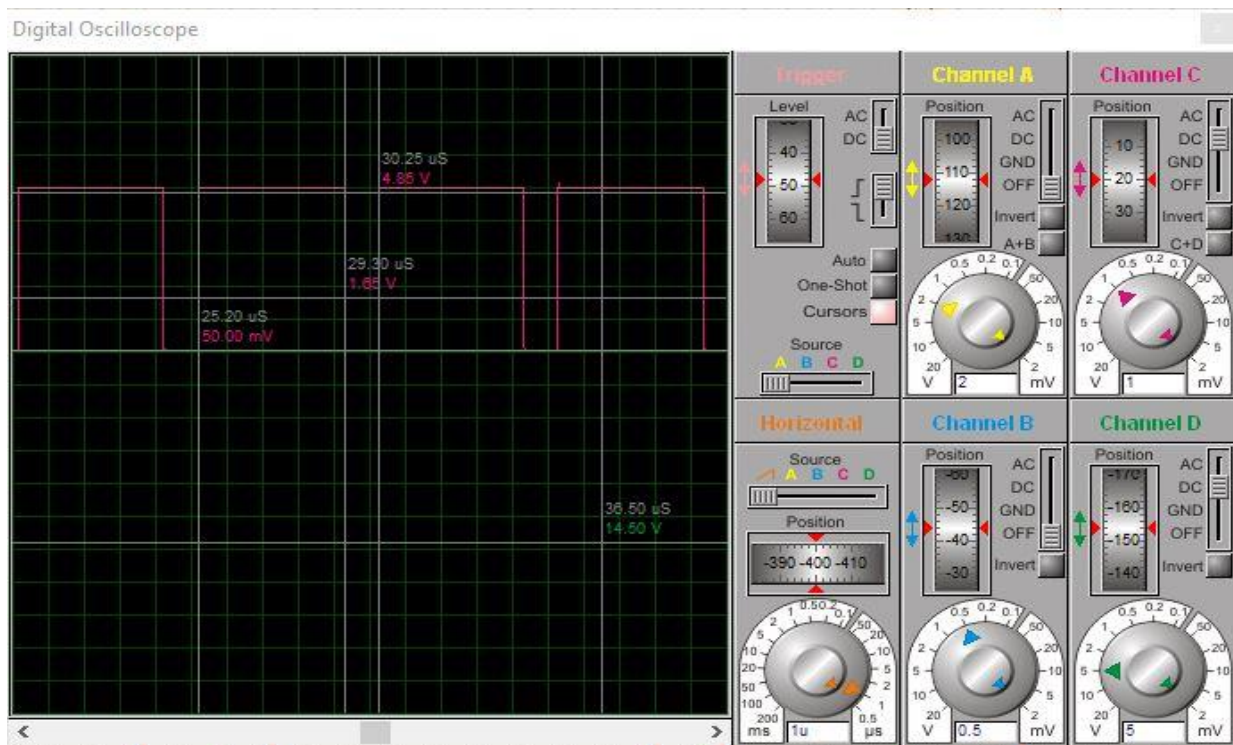


Figure 15 81.18% PWM Duty cycle and 14.4V-14.5V out at 18V (Stage-1)

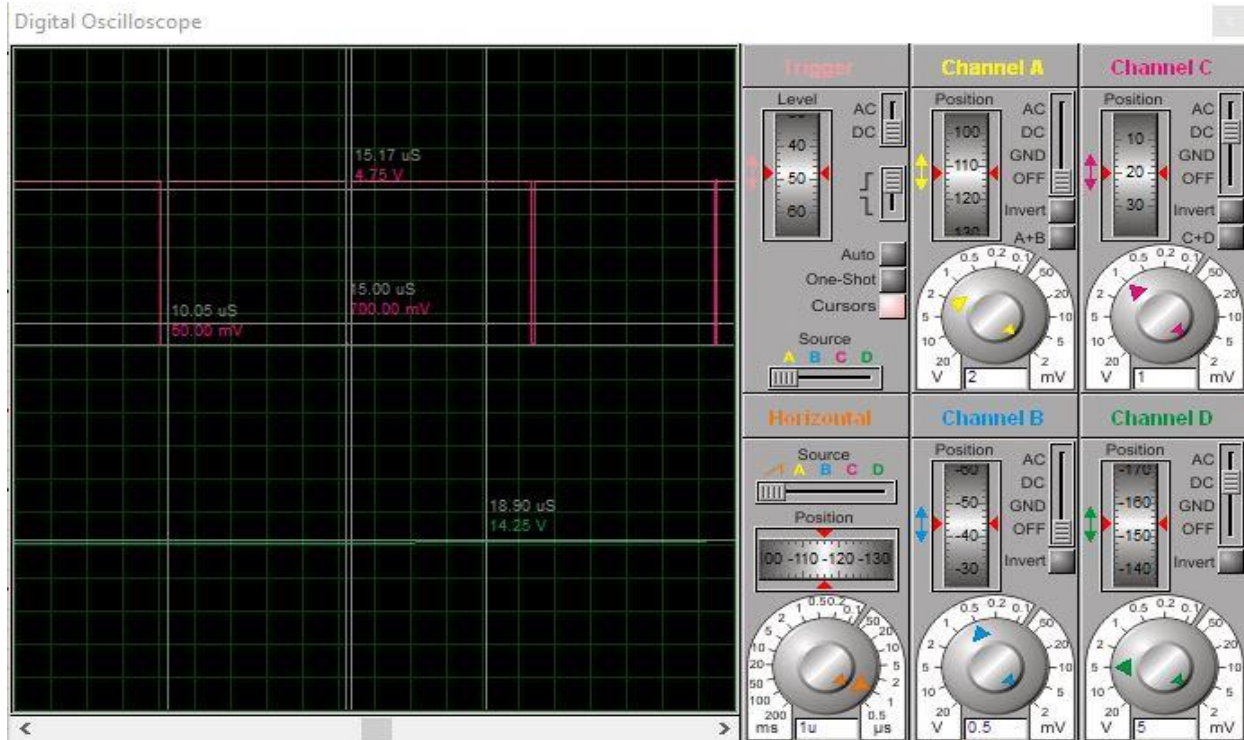


Figure 16 96.68% PWM duty cycle and 15.75V - 13.75V out at 15V (Stage-1)

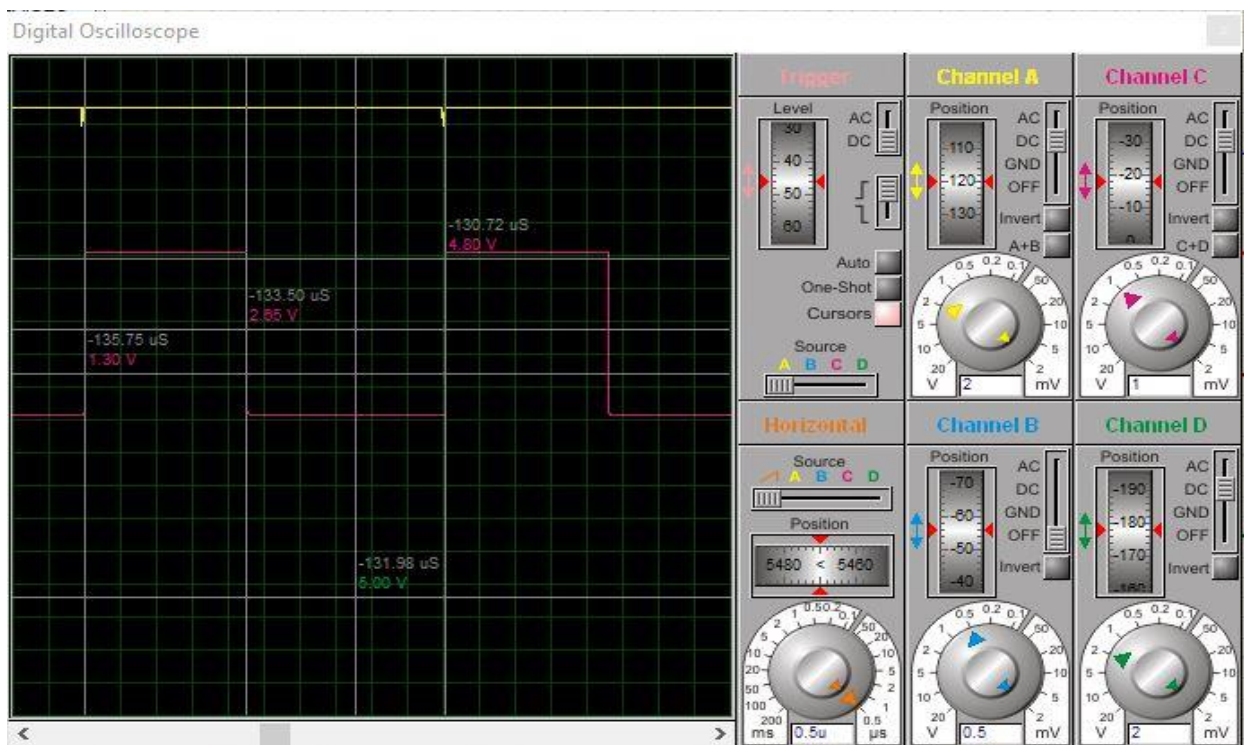


Figure 17 38% PWM Duty cycle and 5.1V-4.9V out at 14.4V (Stage-2)

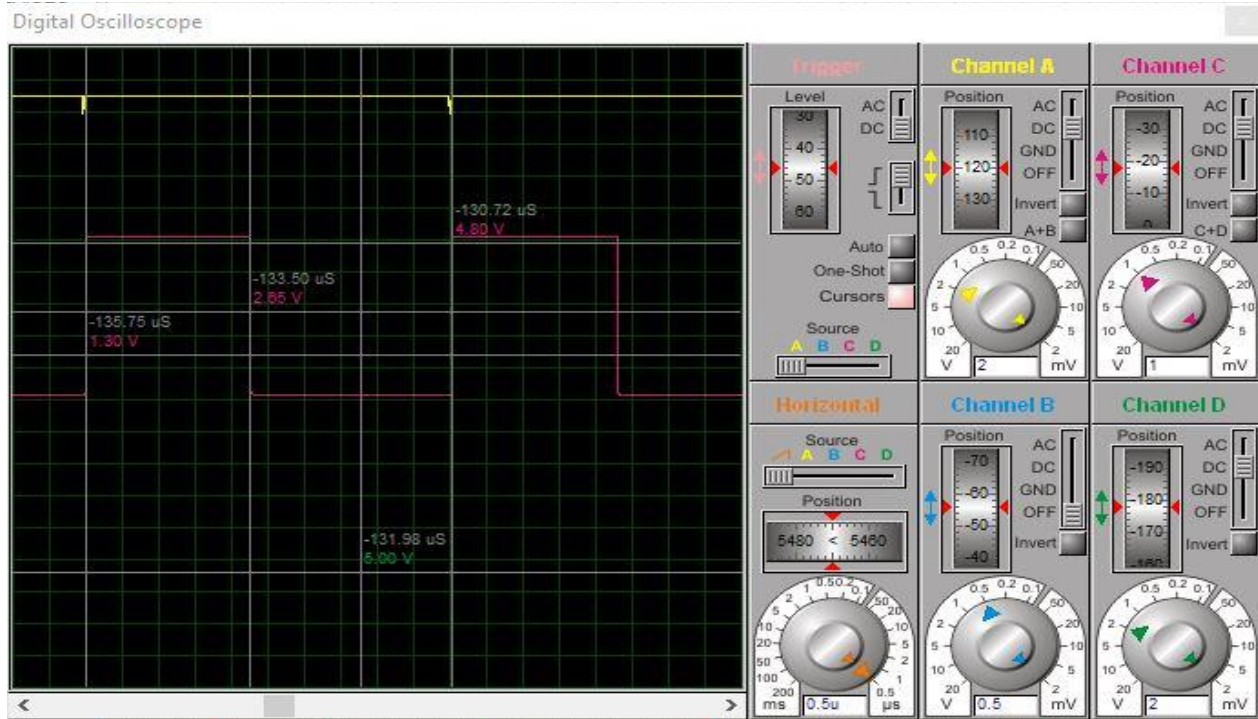


Figure 18 the 44.73% PWM Duty cycle and 5.1V-4.9V out at 12V (Stage-2)

7. Cost Analysis

Table 2 Component Cost Estimation

Component	QTY	Market Price (LKR)	Subtotal (LKR)
NE555	2	20.00	40.00
TL431	2	10.00	20.00
LM311	2	50.00	100.00
TC4420	2	260.00	520.00
IRLZ44N	2	80.00	160.00
1N5822 Schottky	2	10.00	20.00
1N4001	1	5.00	5.00
Resistors	100k	1	1.00
	91k (93k)	1	1.00
	10k (10.5)	1	1.00
	10k	4	4.00
	8.2k	2	2.00
	2.2k	1	1.00
	10k Preset	2	20.00
Capacitors	22uF/50V	1	2.00
	3.3uF/25V	1	2.00
	10nF	2	4.00
	1nF	2	4.00
	560pF (540pF)	2	4.00
Inductors	100uH/3A (85uH)	1	80.00
	33uH (34.5uH)	1	60.00
Estimated Cost			1,051.00

8. Conclusion & Discussion

This project successfully designed and analyzed a two-stage PWM controlled DC–DC buck converter for a portable solar-powered charging system. The primary objective of converting a variable solar panel voltage (15–20 V) into a stable 14.4 V battery charging output and a regulated 5 V, 3 A DC supply was achieved using an analog closed-loop PWM control approach.

The use of a two-stage architecture provided effective voltage regulation and operational flexibility. Stage-1 ensured safe and stable charging of a 12 V lead-acid battery despite variations in solar irradiance, while Stage-2 delivered a well-regulated 5 V output suitable for low-power electronic devices. Simulation results under both 1000 W/m² and 500 W/m² irradiance conditions demonstrated that the system maintained voltage regulation with only minor duty cycle adjustments, confirming the robustness of the closed-loop control strategy.

An important outcome of this work is the successful implementation of analog PWM control using NE555, TL431, LM311, and a TC4420 gate driver, avoiding complex digital controllers or specialized SMPS ICs. This approach improved circuit transparency, ease of understanding, and repairability, making the design especially suitable for educational purposes and low-cost real-world deployment in developing regions such as Sri Lanka.

Efficiency analysis showed that both converter stages exceeded the 85% efficiency requirement, with estimated efficiencies of approximately 98% for Stage-1 and 90% for Stage-2 under midpoint operating conditions. These results confirm that the selected switching frequency (100 kHz), passive component values, and power devices provide a good balance between efficiency, ripple performance, and component size.

Proteus simulations further validated the PWM generation, duty cycle variation, and output voltage regulation across different input voltages. Minor deviations observed at low input voltages (near duty cycle saturation) highlight practical non-idealities such as diode drops, switching losses, and component tolerances, which are expected in real hardware implementations.

Overall, this project demonstrates that a simple, low-cost, analog PWM-controlled two-stage buck converter can effectively meet the requirements of a solar-powered charging system with good efficiency and regulation. Future improvements may include hardware prototyping, thermal analysis, replacement of the diode with synchronous rectification, and integration of MPPT control to further enhance system performance under varying solar conditions.

10. References

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<https://www.alldatasheet.com/datasheet-pdf/pdf/68872/IRF/IRLZ44N.html>

11. Appendix

Appendix A: NE555 Timer Datasheet

Texas Instruments – NE555 Timer

<https://www.ti.com/lit/ds/symlink/ne555.pdf>

Appendix B: TL431 Precision Reference Datasheet

Texas Instruments – TL431

<https://www.ti.com/lit/ds/symlink/tl431.pdf>

Appendix C: LM311 Comparator Datasheet

Texas Instruments – LM311

<https://www.ti.com/lit/ds/symlink/lm311.pdf>

Appendix D: IRLZ44N Power MOSFET Datasheet

Infineon – IRLZ44N MOSFET

<https://www.infineon.com/assets/row/public/documents/24/49/infineon-irlz44n-datasheet-en.pdf>

Appendix E: Power Diode Datasheet

Microchip – Power / Schottky Diode Reference

<https://www1.microchip.com/downloads/en/DeviceDoc/21419D.pdf>